Spatial Context for Wildfire Prediction

Tre Pepper and Tanner Rohloff

The goal of this research-flavor project is to investigate wildfire ignition prediction value change depending on spatial scale, and if these improved predictions justify additional computation cost.

Problem Statement

This Project is a collaboration between Tre Pepper and Tanner Rohloff. It is a Research-flavored project where we try to find whether conditions at multiple spatial scales improves wildfire ignition prediction compared to conditions on a 5 km by 5 km cell. Our wildfire ignitions are determined by prior weather, drought, land cover, seasonal patterns, and historical fire activity. A local cell may omit the predictive information that a broader context will have, such as local vegetation, developed areas, and recent fire activity of the local area. We are trying to see if this broader context provides a predictive value to wildfire ignition across America. We will be using grid-cell and prediction-date pairs for a target's observations, and these will equal one if there is at least one fire greater than 0.5 acres that will occur in the following days, or zero otherwise. The analysis completely depends on fire ignitions, not size, spread, severity, or intensity.

Project Goal and Motivation

Our goal is to build a temporal and special prediction pipeline using PostgreSQL/PostGIS and XGBoost. We are going to determine whether, and for which feature groups, broader context improves prediction beyond a local-only model. We are going to evaluate this empirically, as we do not know if performance will improve from this test. We are going to compare local-only models to local plus 10km, 20km, and 50km models. We additionally are going to investigate whether any improvements justify the additional compute and complexity of the models. Our outputs are going to include risk probability and odds, and a current risk heatmap of the continental United States is a potential application.

Preliminary Methodology

We are going to collect data sets prior to running any models. We are going to align by grid_id and prediction_data. Our predictors will include temperature, daily precipitation, drought severity and history, annual NCLD land-cover percentages, seasonality, historical fire counts, and burned areas over short and multiyear windows. Our weather will come from the closest station with the required observations, and our features must use information available by prediction date. We are additionally going to define our grid geometry before we extract features and will aggregate our variables using multiple value representations. We

Commented [TP1]: at multiple spatial scales improves seven-day wildfire ignition prediction compared with using only conditions within a 5kmx5km cell

Commented [TP2]: Wd will model wildfire ignition risk using

Commented [TP3]: Each observation will be a grid-cell and prediction-date pair. The target will equal one if at least one qualifying wildfire in that grid cell during the following seven days, and zero otherwise. Qualifying fires will be restricted to fires greater than 0.5 acres after excluding intentional and unspecified fire cause categories.

Note: Specifically fires categorized as "Debris and Open Burning," "Recreation and Ceremony," and "Missing data/not specified/undetermined."

Commented [TP4]: spatial

Commented [TP5]: The prediction unit will remain a 5km x 5km cell and we are going…

Note: this wording still doesn't sound right.

Commented [TP6]: In addition to predictive performance, we will record feature-generation and model-training costs for each spatial configuration. This will allow us co compare potential improvements against the…

Commented [TP7]: All prediction observations will be represented by the common key (grid_id, prediction_date).

Commented [TP8]: NLCD

will mainly be using land cover and fire history context for our experiments, with weather, drought, and precipitation being limited for cases.

Our evaluation is going to use non-overlapping chronological periods. We will be using 2000-2016 for training, 2017-2020 for validation, and 2021-2023 for testing. We will be using 2017-2020 to guide our model selection, and our testing will not be done until our final evaluations. Due to wildfire ignition being rare, we will be retaining all positives from training and are sampling around 3 negatives per positive. If we cannot do full population evaluation, we may alter our validation and training to 5 negatives per positive. Sampled probabilities will receive a prevalence adjustment and calibration assessment before being interpreted as real-world risk. Our primary metric will be PR-AUC (Precision Recall- Area Under Curve), which will additionally be supplemented by ROC-AUC (Receiver Operating Characteristic- Area Under Curve), precision, recall, brier score, F1, and log loss.

Milestones

Weeks 1–2: Audit the existing database, confirm source coverage and access, finalize neighborhood definitions, and validate target construction and temporal splits. Weeks 3–4: Extract local predictors, implement negative sampling, and train the 5 km baseline. Weeks 5–6: Build neighborhood features and tune the four spatial configurations. Week 7: Complete held-out evaluation, calibration checks, and ablation studies. Week 8: Consolidate results, document limitations and reproducible procedures, and prepare the final report and a heatmap prototype if feasible

Required Resources

The project requires wildfire records, weather observations, U.S. Drought Monitor data, and annual NLCD layers, accessed through online sources. Our software will include PostgreSql/PostGIS, Python, XGBoost, and data processing and evaluation libraries. A multicore computer will be needed to support the computation for our initial experiments, however, a university computer, or computing resource may be needed. Our limitations include but may not be limited to: the full grid-by-date and multiscale spatial joins may be costly, as well as incomplete or bad data from our sources such as missing station records, incomplete fire histories, and delayed land-cover or fire reporting.

Workload Distribution

Tre Pepper has gathered the data sources and built the database used by the project. Tanner Rohloff will build the data-mining portion, develop the model, and run training and evaluation.

Commented [TP9]: The multi-scale experiments will use landcover and historical fire activity. Weather and precipitation will remain associated with the local 5km prediction cell because the current weather representations is station based rather than a continuous grid.

Commented [TP10]: To prevent information from future periods from entering model training and to better represent the task of predicting future wildfire risk. The test period will not be touched until the final model selection is complete.

Commented [TP11]: As wildfire ignition is a rare event we will be using all qualifying positive observations. Our data will initially use 1:3 positive to negative ratio by randomly sampling grid cell and date pairs.

Note: I'm thinking 1:3 will be good for both based on the size of the data. May be worth increasing or decreasing depending on computational feasibility.

Commented [TP12]: ... because wildfire ignitions are rare relative to all possible grid cell and date observations.

Commented [TP13]: ... will be used for evaluation/ranking

Commented [TP14]: Probably should say what exactly we are using this for.

to construct the spatial grid, store source and processed relations, generate spatial neighborhood, perform joins, and make the datasets used for ML.

Commented [TP15]: Maybe add related works section: https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025EF006935

Commented [TP16]: Name the datasets more explicitly